

Assignment_3_cwood19

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0: Partition the data into training and validation sets 60/40 Fields: Age, Experience, Income, Family, CCAvg, Mortgage, Securities.Account, CD.Account, Online, CreditCard, Education, Personal.Loan

```
# Load packages
```

```
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(fastDummies)
```

```
## Warning: package 'fastDummies' was built under R version 4.5.3
```

```
# Load dataset
```

```
UniversalBank <- read.csv("UniversalBank.csv")
```

```
# Remove unused variables
```

```
UniversalBank <- UniversalBank[, !(names(UniversalBank) %in% c("ID", "ZIP.Code"))]
```

```
# Handle extra Education variables with dummies
```

```
UniversalBank <- dummy_cols(  
  UniversalBank,  
  select_columns = "Education",  
  remove_first_dummy = FALSE,  
  remove_selected_columns = TRUE  
)
```

```
# Set reproducibility
```

```
set.seed(123)
```

```
# Train the model for 60% / p = .6; y = target variable
```

```
train_index <- createDataPartition(  
  UniversalBank$Personal.Loan,  
  p = 0.60,  
  list = FALSE  
)
```

```
train_data <- UniversalBank[train_index, ]
```

```
valid_data <- UniversalBank[-train_index, ]
```

A. Create a pivot table for the training data with Online as a column variable, CreditCard as a row variable, and Personal.Loan as a secondary row variable. The values inside the table should convey the count. In R use functions melt() and cast(), or function table().

```
# Generate pivot table with table()
pivot_table <- table(
  CreditCard = train_data$CreditCard,
  Personal.Loan = train_data$Personal.Loan,
  Online = train_data$Online
)

ftable(pivot_table)
```

```
##               Online    0    1
## CreditCard Personal.Loan
## 0           0           785 1145
##           1           65  122
## 1           0          317  475
##           1           34   57
```

A) The table identifies that more people are using online banking than not, and those that use online banking more frequently accept loans.

B. Consider the task of classifying a customer who owns a bank credit card and is actively using online banking services. Looking at the pivot table, what is the probability that this customer will accept the Personal.Loan offer? [This is the probability of Personal.Loan acceptance (Personal.Loan = 1) conditional on having a bank credit card (CreditCard = 1) and being an active user of online banking services (Online = 1)].

```
# P(Personal.Loan = 1 / CreditCard = 1, Online = 1)
loan_yes <- pivot_table["1", "1", "1"]

total_cc_online <- sum(pivot_table["1", , "1"])

prob_pivot <- loan_yes / total_cc_online

prob_pivot
```

```
## [1] 0.1071429
```

B) Based on the condition of having a CC and being an online service user, there's an estimated 10.7% chance of accepting a loan.

C. Create two separate pivot tables for the training data. One will have Personal.Loan (rows) as a function of Online (columns) and the other will have Personal.Loan (rows) as a function of CreditCard.

```
loan_online <- table(
  Personal.Loan = train_data$Personal.Loan,
  Online = train_data$Online
)

loan_online
```

```
##           Online
## Personal.Loan    0    1
##           0 1102 1620
##           1   99  179
```

```
loan_cc <- table(
  Personal.Loan = train_data$Personal.Loan,
  CreditCard = train_data$CreditCard
)

loan_cc
```

```
##           CreditCard
## Personal.Loan    0    1
##           0 1930  792
##           1  187   91
```

D. Compute the following quantities [$P(A \mid B)$ means “the probability of A given B”]: 1. $P(\text{CreditCard} = 1 \mid \text{Personal.Loan} = 1)$ (the proportion of credit card holders among the loan acceptors) 2. $P(\text{Online} = 1 \mid \text{Personal.Loan} = 1)$ 3. $P(\text{Personal.Loan} = 1)$ (the proportion of loan acceptors) 4. $P(\text{CreditCard} = 1 \mid \text{Personal.Loan} = 0)$ 5. $P(\text{Online} = 1 \mid \text{Personal.Loan} = 0)$ 6. $P(\text{Personal.Loan} = 0)$

```
### D.1 ###
p_cc_given_loan1 <-
  loan_cc["1","1"] / sum(loan_cc["1",])

p_cc_given_loan1
```

```
## [1] 0.3273381
```

```
### D.2 ###
p_online_given_loan1 <-
  loan_online["1","1"] / sum(loan_online["1",])

p_online_given_loan1
```

```
## [1] 0.6438849
```

```
### D.3 ###
p_loan1 <-
  sum(train_data$Personal.Loan == 1) /
  nrow(train_data)

p_loan1
```

```
## [1] 0.09266667
```

```
### D.4 ###
p_cc_given_loan0 <-
  loan_cc["0","1"] / sum(loan_cc["0",])

p_cc_given_loan0
```

```
## [1] 0.2909625
```

```
### D.5 ###
p_online_given_loan0 <-
  loan_online["0","1"] / sum(loan_online["0",])

p_online_given_loan0
```

```
## [1] 0.5951506
```

```
### D.6 ###
p_loan0 <-
  sum(train_data$Personal.Loan == 0) /
  nrow(train_data)

p_loan0
```

```
## [1] 0.9073333
```

E. Use the quantities computed above to compute the naive Bayes probability $P(\text{Personal.Loan} = 1 \mid \text{CreditCard} = 1, \text{Online} = 1)$.

```
# Calculate numerator:  $P(CC = 1 \mid L = 1) P(O = 1 \mid L = 1) P(L = 1)$ 
numerator <-
  p_cc_given_loan1 *
  p_online_given_loan1 *
  p_loan1

# Calculate denominator:  $P(CC = 1 \mid L = 1) P(O = 1 \mid L = 1) P(L = 1) + P(CC = 1 \mid L = 0) P(O = 1 \mid L = 0) P(L = 1)$ 
denominator <-
  numerator +
  p_cc_given_loan0 *
  p_online_given_loan0 *
  p_loan0

# Calculate  $P(L = 1 \mid CC = 1, O = 1) = \text{numerator} / \text{denominator}$ 
nb_prob <- numerator / denominator

nb_prob
```

```
## [1] 0.1105637
```

NB probability computed at 0.1105637 with current probability $P(\text{Loan} = 1 \mid \text{CC} = 1, \text{Online} = 1)$

F. Compare this value with the one obtained from the pivot table in (B). Which is a more accurate estimate?

```
# Generate values
prob_pivot
```

```
## [1] 0.1071429
```

```
nb_prob
```

```
## [1] 0.1105637
```

F) Difference between `prob_pivot` & `nb_prob` = .0034208 or .34%, which is relatively insignificant, but the pivot table is closer to 1 and computes directly from the observed data rather than NB making assumptions about conditional independence between CC & Online given Loan.

G. Which of the entries in this table are needed for computing $P(\text{Personal.Loan} = 1 \mid \text{CreditCard} = 1, \text{Online} = 1)$? Run naive Bayes on the data. Examine the model output on training data, and find the entry that corresponds to $P(\text{Personal.Loan} = 1 \mid \text{CreditCard} = 1, \text{Online} = 1)$. Compare this to the number you obtained in (E)

```
# Add library for naiveBayes function  
library(e1071)
```

```
##  
## Attaching package: 'e1071'  
  
## The following object is masked from 'package:ggplot2':  
##  
##     element
```

```
# Fit the model  
nb_model <- naiveBayes(  
  as.factor(Personal.Loan) ~ CreditCard + Online,  
  data = train_data  
)  
  
nb_model
```

```
##  
## Naive Bayes Classifier for Discrete Predictors  
##  
## Call:  
## naiveBayes.default(x = X, y = Y, laplace = laplace)  
##  
## A-priori probabilities:  
## Y  
##           0           1  
## 0.90733333 0.09266667  
##  
## Conditional probabilities:  
##   CreditCard  
## Y      [,1]      [,2]  
## 0 0.2909625 0.4542897  
## 1 0.3273381 0.4700881  
##  
##   Online  
## Y      [,1]      [,2]  
## 0 0.5951506 0.4909531  
## 1 0.6438849 0.4797134
```

```
# Display the model entries
nb_model$tables$CreditCard
```

```
##      CreditCard
## Y      [,1]      [,2]
## 0 0.2909625 0.4542897
## 1 0.3273381 0.4700881
```

```
nb_model$tables$Online
```

```
##      Online
## Y      [,1]      [,2]
## 0 0.5951506 0.4909531
## 1 0.6438849 0.4797134
```

NB model uses the prior probabilities $P(\text{Personal.Loan} = 1) = .0927$ and $P(\text{Personal.Loan} = 0) = .9073$ with the conditional probabilities (.3273, .6439, .2910, and .5952) to yield the results for $P(\text{Personal.Loan} = 1 \mid \text{CreditCard} = 1, \text{Online} = 1) = .1106$, which very closely aligns to the pivot table's evaluation at .1071.